Rishi More

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Summary

ML systems engineer and Johns Hopkins CS graduate focused on LLM inference, CUDA/Triton kernels, quantization, and adaptive serving. **ICML 2026 co-first author** with vLLM integration, Nsight profiling, and correctness-gated benchmarks spanning weight bandwidth, KV-cache memory, and speculative decoding.

Experience

InterSystems **July 2026 – September 2026**
Core Development Program, AI Experience *Boston, Massachusetts*

- Shipped **LLM-backed clinical features** for TrakCare, an electronic health record platform deployed across **29 countries**.
- Integrated an **AI-generated patient summary column** into clinician workflows, consolidating patient histories into one overview to simplify chart review and reduce navigation across clinical records.

Johns Hopkins University Data Science and AI Institute (DSAI) **May 2025 – June 2026**
Graduate Research Assistant, LLM Inference Optimization *Baltimore, Maryland*

- Co-first authored *Conformal Thinking* (**ICML 2026 main track**); designed **dual-threshold stopping** to exit on confident answers or stalled reasoning, reducing reasoning tokens by **12% at matched accuracy**.
- Built a calibration and evaluation pipeline using vLLM across **4 reasoning models, 4 benchmarks, and 2.7K+ problems**, capturing intermediate answers, confidence trajectories, and token counts to compare stopping policies.
- Implemented **UCB-calibrated signal-threshold selection**, choosing the lowest-token policy satisfying a user-set risk budget; benchmarked against naive threshold tuning and stress-tested false-positive and premature-stop risk under dataset and response-length shifts.

Mastek **November 2022 – July 2023**
Machine Learning Engineer Intern *Mumbai, India*

- Deployed an on-premises edge voice-authentication system with **<100 ms p99 latency** and **96.6% accuracy**, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment.
- Built a TensorFlow pipeline over **1M+ voice samples**; spectral-gating denoising, augmentation, and robust feature extraction improved authentication reliability by **20%** in high-noise conditions.

Projects

KV Roofline: INT4 KV Cache & Fused Decode Attention | *Triton, PyTorch, CUDA*

- Implemented **per-channel K / per-token V** quantization, packed INT4 storage, and an FP16 residual window; reduced measured KV storage **3.47**× at 4K tokens on Qwen2.5-1.5B (**28.7 to 8.3 KB/token**).
- Built Triton dequant-attention with token-axis splits and online softmax; measured **7.54 to 0.55 ms per layer at 32K** versus PyTorch FP16 GQA SDPA math path on an RTX 4070 Laptop.
- Gated kernels against a dequantized oracle at ≥**0.999 cosine**; separately checked model quality through an HF-compatible cache, observing **+1.0% perplexity** on a 1,023-token WikiText slice.

Conformal Serve: Risk-Controlled Adaptive LLM Serving | *vLLM, FastAPI, Docker Compose, Prometheus*

- Built an engine-agnostic adaptive-stopping core shared by deterministic mock-engine CI and vLLM's AsyncLLMEngine; achieved **100% policy branch coverage** with `mypy --strict` and exposed client-set risk α through an OpenAI-compatible API.
- On Qwen2.5-1.5B-Instruct and an 8 GB RTX 4070, reduced reasoning tokens **64.7% at** $\alpha = 0.10$ and **71.4% at** $\alpha = 0.30$ while maintaining realized error $\leq \alpha$ on small held-out arithmetic sets.
- Calibrated per-α thresholds offline by distribution-free risk control into immutable, content-addressed artifacts; shipped Docker Compose with Prometheus/Grafana and a one-command benchmark reporting **95%** Clopper–Pearson error bounds.

Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | *CUDA C++, Nsight*

- Profiled Qwen2.5-1.5B batch-1 decode with Nsight; attributed ∼**81% of GPU-kernel time** to weight GEMVs and measured dominant MLP projections at ∼**247 GB/s** against a ∼**250 GB/s** DRAM ceiling.
- Built a correctness-gated fused CUDA INT4 group-dequantization + GEMV kernel reaching ∼**223 GB/s**; accelerated batch-1 MLP projections by **2.38**× versus FP16 GEMM and mapped the crossover to batch 2 as weight reuse made GEMM faster.

Education

Johns Hopkins University **August 2024 – May 2026**
M.S.E. in Computer Science *GPA: 3.9/4.0*

University of Mumbai **July 2020 – July 2024**
B.Tech. in Computer Science, Honors in AI/ML *CGPA: 9.85/10*

Technical Skills

Languages: Python, C++, CUDA C++, C, SQL, Bash
ML & Serving: PyTorch, Triton, vLLM, Hugging Face Transformers, TensorFlow, FastAPI
Performance: Nsight Compute/Systems, roofline analysis, kernel fusion, INT4/INT8 quantization, KV caching, speculative decoding
Infrastructure & Methods: Linux, Docker, GitHub Actions, AWS, pytest, conformal risk control, calibration, LLM evaluation